TelecomInsight BW: Telecommunications Service Quality and Customer Experience Analytics Platform

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October 15, 2026

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Chapter 1

Literature Review

1.1 Introduction and Approach

TelecomInsight BW is a system that organizes and categorizes telecommunications complaints in Botswana, analyses them to identify common problems and trends, and presents the results through an interactive dashboard.

This review is organized around the features of existing systems rather than around studies in isolation. It asks four questions in order: what systems and projects already exist in this area; what features do they provide; which of those features are useful to TelecomInsight BW and how were they implemented or adapted; and what additional features does TelecomInsight BW still need in order to address problems that existing systems do not solve. This structure was chosen so that each source can be linked to a specific design decision in the system.

1.2 Existing Projects and Systems

Four categories of existing work were reviewed: operator complaint systems in Botswana, the national regulator's complaint system, regulator dashboards in other countries, and research prototypes that analyse complaints or network data. The table below summarizes what each does.

System / Study	Type	What It Does
Mascom complaints procedure	Operator system (Botswana)	Lets customers lodge a complaint directly with the operator and escalates unresolved cases from a departmental manager to a senior manager within set response times.

System / Study	Type	What It Does
BOCRA complaints handling	Regulator system (Botswana)	Investigates complaints escalated from operators, and reports on recurring national complaint themes such as billing, missing airtime and data bundles, faulty lines and slow internet speeds, broken down by operator.
Ofcom complaints data (UK)	Regulator dashboard	Publishes quarterly complaint volumes per 100,000 customers for each major provider, shown through interactive online charts with downloadable data.
FCC Consumer Complaints Data Center (USA)	Regulator dashboard	Publishes individual self-reported complaint records updated close to daily, with charts, maps and an API for filtering and analysis.
Choi (2018); Rocha Junior et al. (2023); Ibrahim (2026)	Research prototypes	Demonstrate that telecommunications complaints can be automatically classified into categories and that future complaint volumes can be forecast using machine learning.
Baddour et al. (2025)	Research prototype	Applies big-data analytics to network-performance measurements to support evaluation of network quality.
Neri et al. (2025); Jarske et al. (2023)	Research (visualization)	Show that effective visualization improves understanding of analytical results, and that dashboards should make the origin of displayed information clear.

Two observations follow from this. First, complaint handling already exists in Botswana at both operator and regulator level, so TelecomInsight BW is not introducing complaint collection for the first time. Second, the aggregation and visualization of complaint data is well established internationally, but not in the Botswana market.

1.3 What Features Do These Systems Have?

Comparing the reviewed systems by feature rather than by description makes the differences between them clearer. The table below sets out the main features and shows which systems provide them.

Feature	Mascom	BOCRA	Ofcom	FCC	TelecomInsi BW
Record individual complaints	Yes	Yes	No	Yes	Yes
Categories complaints by issue type	Internal	Yes	Yes	Yes	Yes
Compare multiple providers	No	Yes	Yes	Yes	Yes
Show trends over time	No	Partly	Yes	Yes	Yes
Geographic / location view	No	No	No	Yes	Yes
Interactive filters and charts	No	No	Yes	Yes	Yes
Downloadable / reusable data	No	No	Yes	Yes	Yes
Linked to network measurements	No	No	No	No	Yes
Botswana-specific	Yes	Yes	No	No	Yes

The comparison shows a clear pattern. The Botswana systems are strong on local relevance but weak on analysis and presentation: Mascom handles cases within one company, and BOCRA categorizes and compares providers but releases findings as periodic written reports. The international dashboards are the reverse: Ofcom and the FCC provide strong interactive analysis, trends and public data access, but are built for their own markets. No reviewed system links complaints to network measurements.

1.4 Which Features Are Useful to Us, and How We Implemented Them

Not every feature of the reviewed systems was relevant to this project. Formal case-escalation workflows, for example, are central to the Mascom and BOCRA systems because those organizations must resolve individual disputes, but TelecomInsight BW is an analysis tool rather than a complaint-resolution service and therefore does not reproduce them. The features that were adopted, and how each was implemented or adapted, are given in the table below.

Feature Adopted	**Where We Saw It**	**How We Implemented or Adapted It**
Complaint categorization	BOCRA reporting; Choi (2018); Rocha Junior et al. (2023)	Complaints are stored against fixed issue categories (for example network coverage, slow data speed, billing, dropped calls) chosen to match the problem types BOCRA already reports for Botswana, so that records can be counted and compared instead of read one by one.
Provider-level comparison	BOCRA reports; Ofcom complaints data	The dashboard groups complaints by provider so that problem patterns can be compared across BTC, Mascom and Orange, rather than being limited to one operator as in the Mascom system.
Trend analysis over time	Ofcom quarterly series; FCC data center	Each complaint carries a date, allowing the dashboard to chart complaint volumes by month and show whether a particular issue is rising or falling.
Interactive filtering and charts	Ofcom and FCC dashboards; Neri et al. (2025)	The dashboard uses charts with filters for provider, location, issue type and date, so users can narrow the view themselves instead of reading a fixed report.
Geographic view	FCC complaint maps	Complaints are recorded against a location so that areas with repeated problems can be identified, which is not offered by the local systems reviewed.
Clear data origin	Jarske et al. (2023)	The system records the source of each record (survey response, public report or network measurement) and applies cleaning rules before analysis, so displayed figures can be traced back.

In each case the feature was adapted rather than copied. Ofcom reports complaints per 100,000 customers because reliable subscriber figures are published for each UK provider; comparable figures are not readily available for Botswana operators, so TelecomInsight BW compares complaint counts and proportions by category instead. Similarly, the automated classification used by Choi (2018) and Rocha Junior et al. (2023) was not implemented directly, because the initial dataset is small and manually categorized; instead the data is stored in a structured form that would allow such methods to be added later.

1.5 What Additional Features Does TelecomInsight BW Need?

The feature comparison also identifies gaps that no reviewed system fills. These gaps define the additional features that TelecomInsight BW must provide, sumed up in the table below.

Additional Feature	**Why It Is Needed**	**Not Provided By**
Combining customer complaints with network measurements	A network can appear technically available while users still experience slow or unreliable service; combining both shows whether reported problems match measurable conditions.	Mascom, BOCRA, Ofcom, FCC — all handle complaints only.
Directly collected customer-experience data	Only complaints that customers choose to escalate reach BOCRA, so many everyday problems are never recorded. Survey responses capture these.	All reviewed systems, which rely on submitted complaints.
Botswana-focused interactive dashboard	Ofcom and FCC show the value of interactive tools, but their categories, providers and geography are not applicable to Botswana.	Ofcom and FCC (foreign markets); BOCRA (static reports).
Structured data prepared for future automated analysis	Research shows complaints can be classified and forecast automatically; storing clean, categorized data now makes this possible later.	Local operator and regulator systems, which store cases for resolution rather than analysis.

The most important of these is the combination of customer complaints with network measurements. The reviewed complaint systems record what customers report, while the network-analytics research reviewed by Baddour et al. (2025) analyses technical measurements, but the two are treated separately. Bringing them together allows the system to indicate whether reported problems correspond to measurable network conditions in a particular area.

1.6 Research Gap

Existing operator and regulator systems in Botswana collect and categorise complaints but present them mainly for case resolution and periodic reporting. Regulator dashboards in other countries demonstrate interactive aggregation, trend analysis and public data access, but are designed for different markets and do not incorporate network measurements. Research studies show that complaints can be classified and forecast automatically and that network data can be analyzed at scale, but they address these problems separately rather than in one tool.

The gap is therefore not in any single feature, but in the combination: no reviewed system brings together locally collected customer feedback, complaint categorization, provider comparison, geographic and time-based analysis, network measurements and interactive visualization in one platform designed for the Botswana telecommunications environment. TelecomInsight BW is designed to address this combination.

1.7 Summary

Reviewing existing systems by feature produced three outcomes for this project. It confirmed which features are already proven and worth adopting, including complaint categorization, provider comparison, trend analysis and interactive filtering. It identified which features were not appropriate for this project, such as formal case-escalation workflows and per-100,000-customer normalization. It also showed which features must be added, principally the linking of complaints to network measurements, the direct collection of customer-experience data through surveys, and an interactive dashboard built specifically for the Botswana market.

These findings guided the decisions about what data TelecomInsight BW collects, how that data is categorized and prepared, and how the results are presented, and they provide the basis for the system design described in the following chapter.

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